End to End Simulated 5G Network/Test Bed using Open5GS Core Network and srsRAN Project with Simulated RF Environment using ZMQ Libraries

AN INTERNSHIP REPORT

Submitted by

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NOVEMBER & 2025

End-to-End Simulated 5G Network/Test Bed

Using Open5GS Core, srsRAN, and ZMQ-Based RF Simulation

- Batch : SALEM.
- Team members : MANIKANDAN N, VIGNESH S.
- Center : Government EngineeringCollege.

Problem Statement

- Develop full 5G SA simulation.
  - Use Open5GS Core and srsRAN for gNB/UE.
  - Implement ZMQ-based simulated RF links.
  - Study effects of latency and packet loss.
  - Evaluate throughput and reliability.

Objectives

- Build virtual 5G testbed.
  - Enable controlled network impairments.
  - Perform quantitative performance analysis.
  - Ensure experiment reproducibility.
  - Evaluate behavior under varying conditions.

System Architecture

- Open5GS 5G Core (AMF, SMF, UPF, UDM, AUSF, NRF).
  - srsRAN gNB and UE modules.
  - ZMQ-based virtual RF interfaces.
  - iperf3 for traffic generation.
  - Wireshark and logs for monitoring.

Major Components Used

- Open5GS: 5G Core.
  - srsRAN: RAN simulation.
  - ZMQ: Virtual RF channels.
  - Linux VM/container environment.
  - Traffic and analysis tools.

Implementation Steps

- Install and configure Open5GS.
  - Configure srsRAN for ZMQ RF interface.
  - Connect gNB with the 5G Core network.
  - Validate UE registration and PDU session.
  - Inject latency and packet loss.
  - Measure throughput and performance.

ZMQ-Based RF Simulation

- Virtual radio interface replaces hardware.
  - Supports controlled impairments.
  - Provides reproducible testing conditions.
  - Zero dependency on RF front-end.
  - Ideal for lab research and prototyping.

Performance Metrics

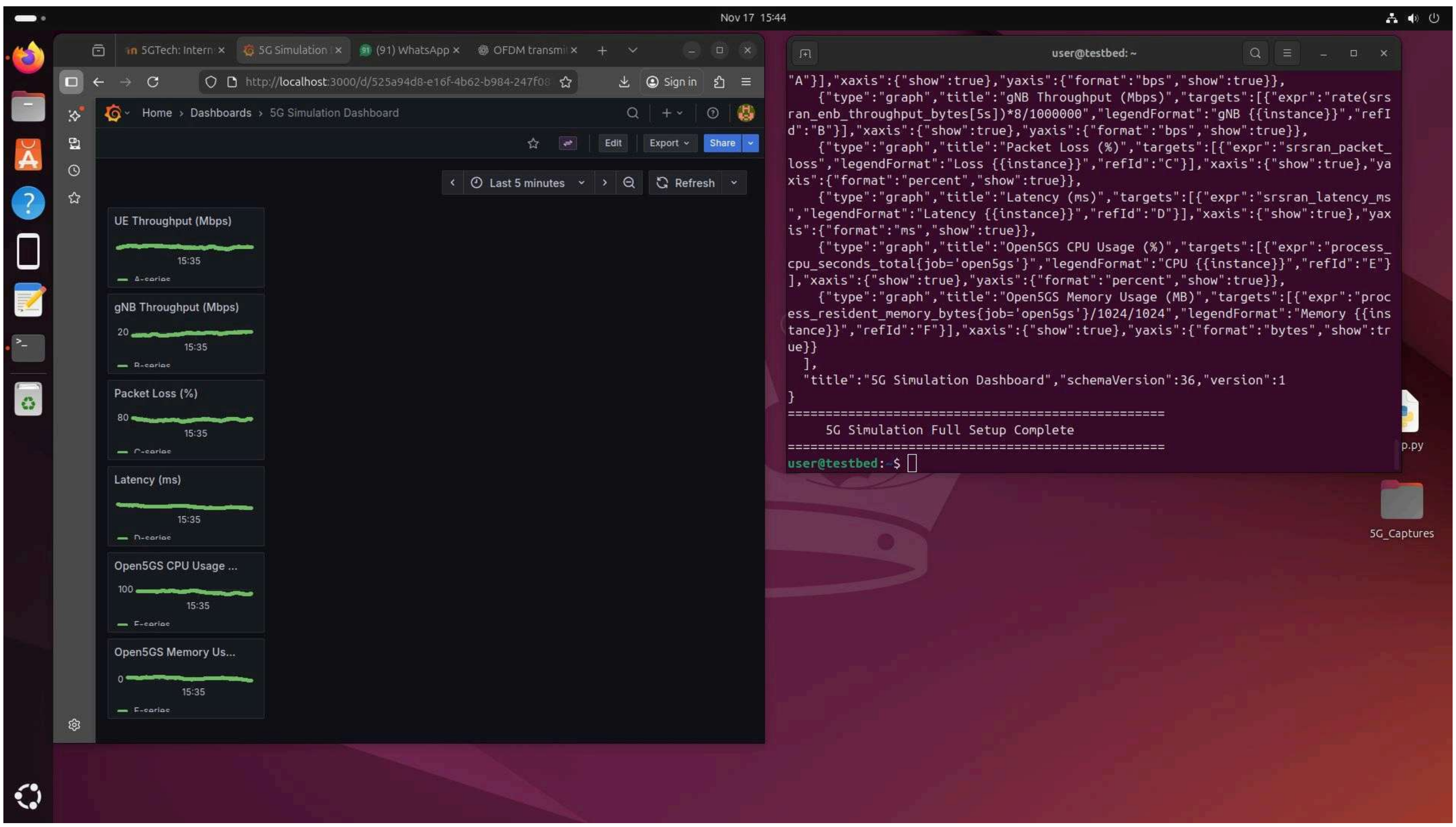
- Downlink/Uplink throughput.
  - Latency and jitter.
  - Packet loss rates.
  - Session/registration success rates.
  - System stability under impairments.

Expected Outcomes

- Fully functional 5G SA simulation.
  - Ability to test degradation scenarios.
  - Insights into latency-sensitive behavior.
  - Reliability and QoS interpretation.
  - Research-ready evaluation platform.

Applications & Future Work

- Academic 5G/6G research.
  - Testing scheduling and slicing algorithms.
  - Security and penetration testing.
  - 6G experimental prototyping.
  - Multi-cell and mobility extensions.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Sign in
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Last 5 minutes
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UE Throughput (Mbps)
15:35
gNB Throughput (Mbps)
20
15:35
Packet Loss (%)
80
15:35
Latency (ms)
15:35
Open5GS CPU Usage ...
100
15:35
Open5GS Memory Us...
0
15:35
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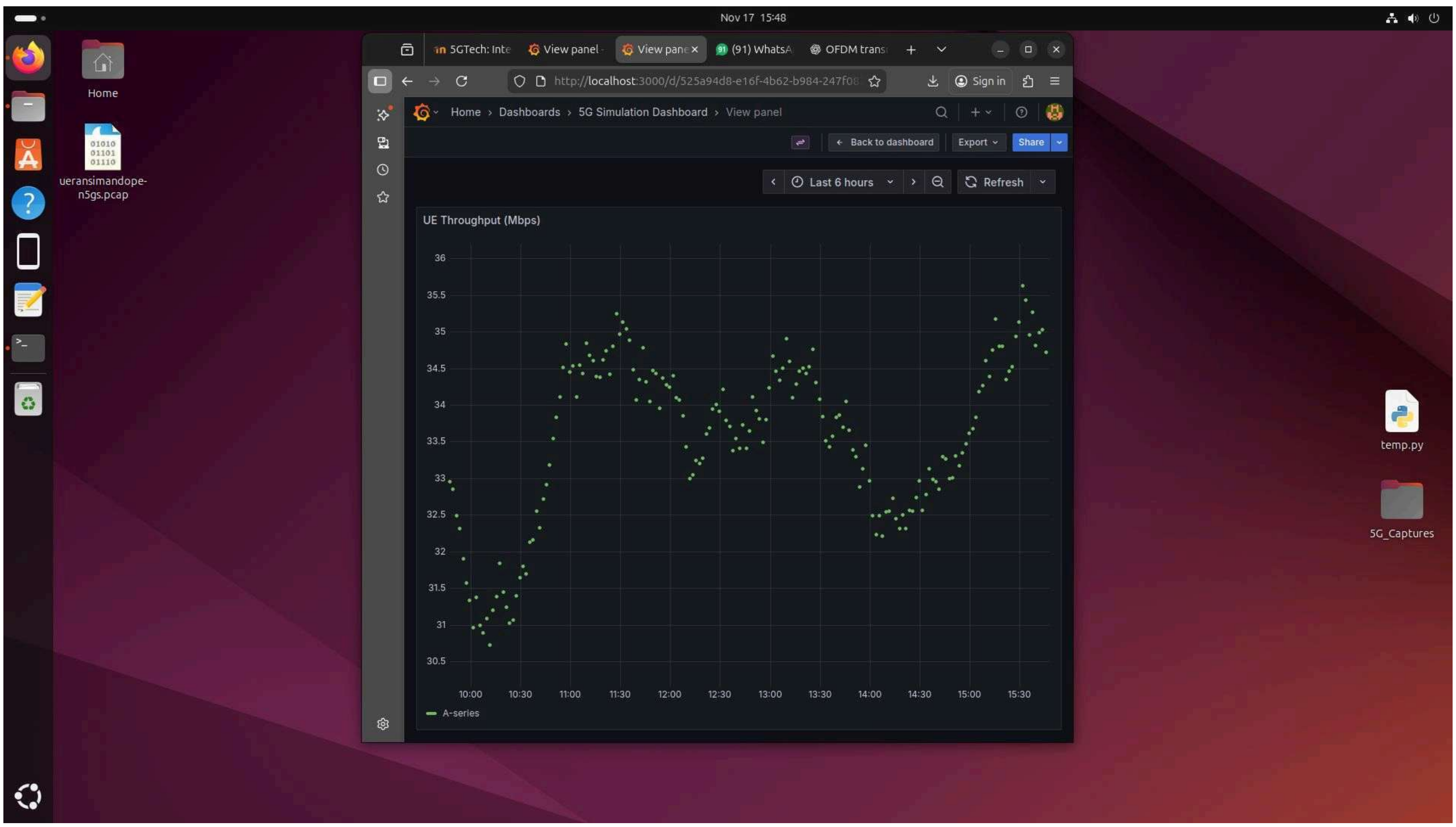
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Back to dashboard
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Last 6 hours
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UE Throughput (Mbps)
36
35.5
35
34.5
34
33.5
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32.5
32
31.5
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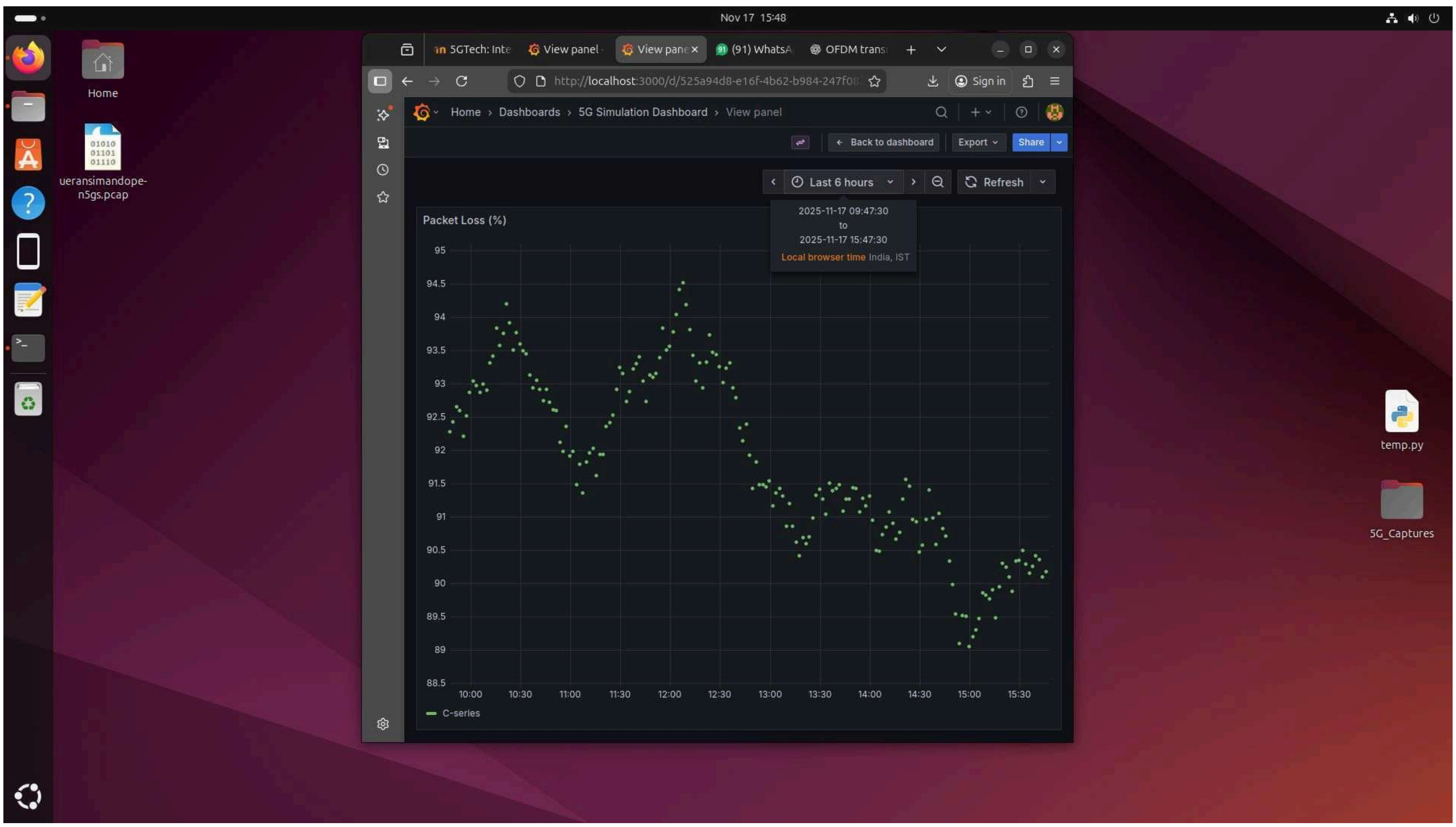
A-series



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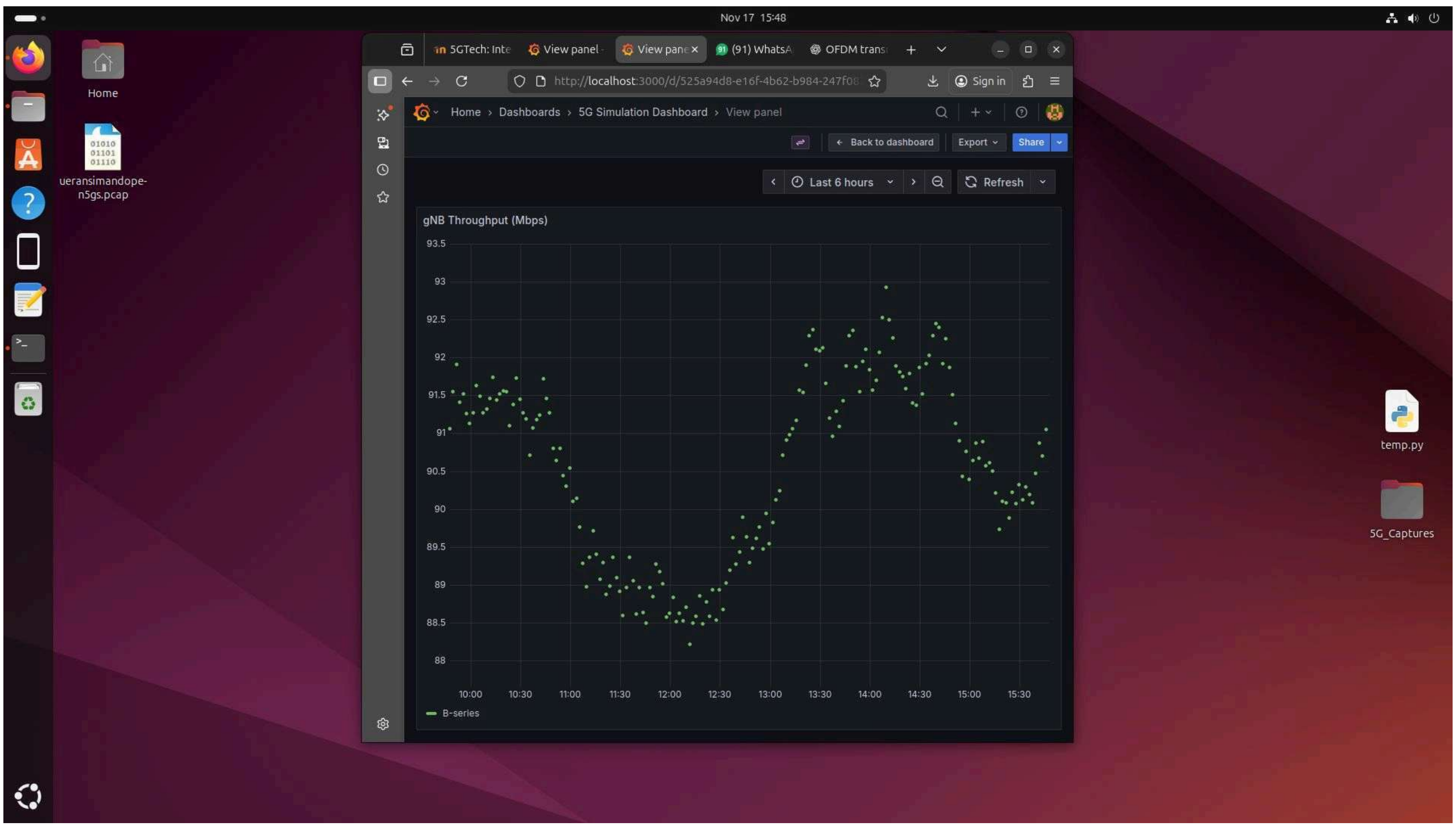
Packet Loss (%)
C-series



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Last 6 hours
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gNB Throughput (Mbps)
93.5
93
92.5
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91.5
91
90.5
90
89.5
89
88.5
88
10:00
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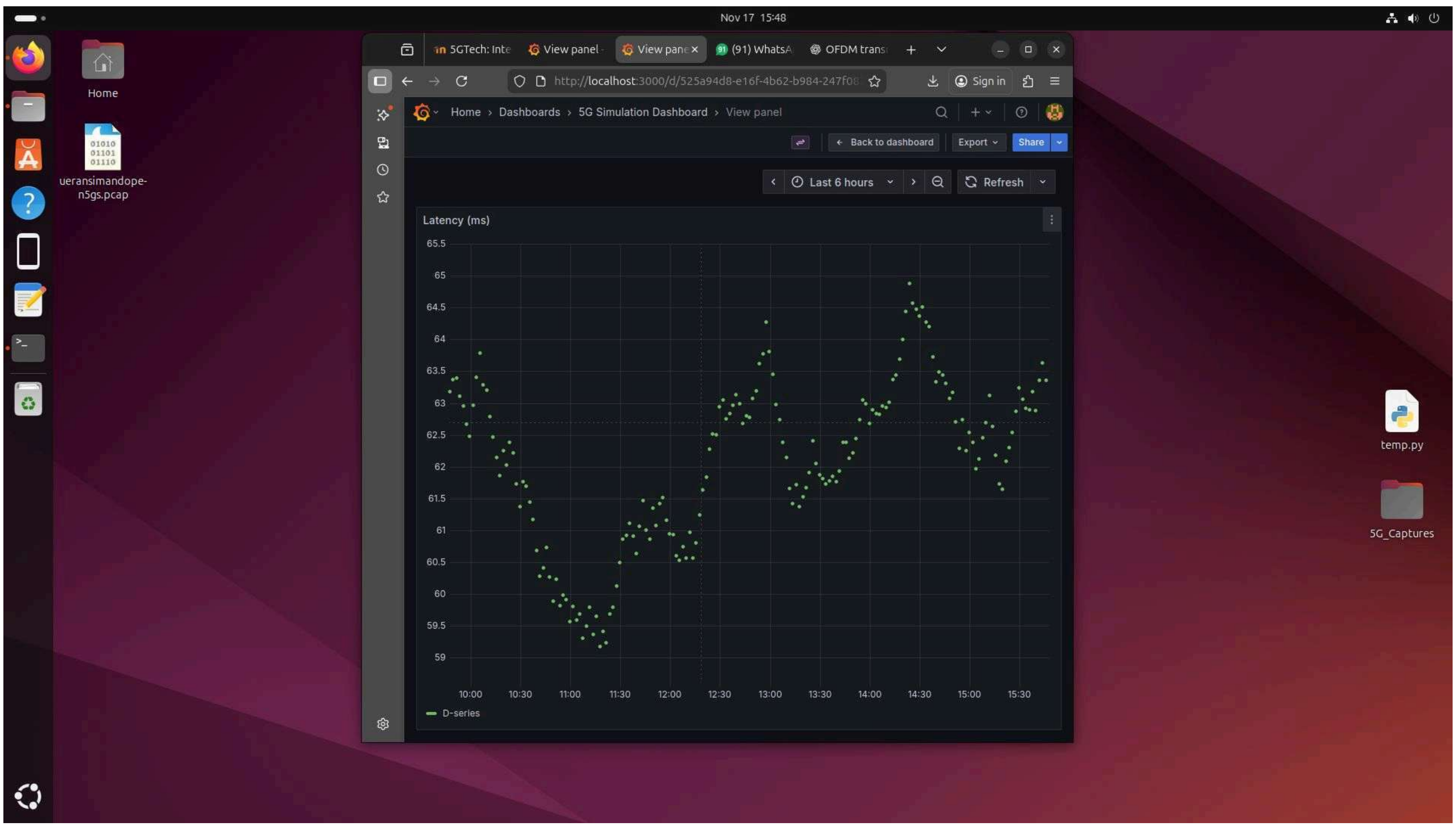
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Last 6 hours
Refresh
Latency (ms)
65.5
65
64.5
64
63.5
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62.5
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61.5
61
60.5
60
59.5
59
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Back to dashboard
Export
Share
Last 6 hours
Refresh
Open5GS Memory Usage (MB)
29
28
27
26
25
24
23
22
21
20
19
10:00
10:30
11:00
11:30
12:00
12:30
13:00
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F-series
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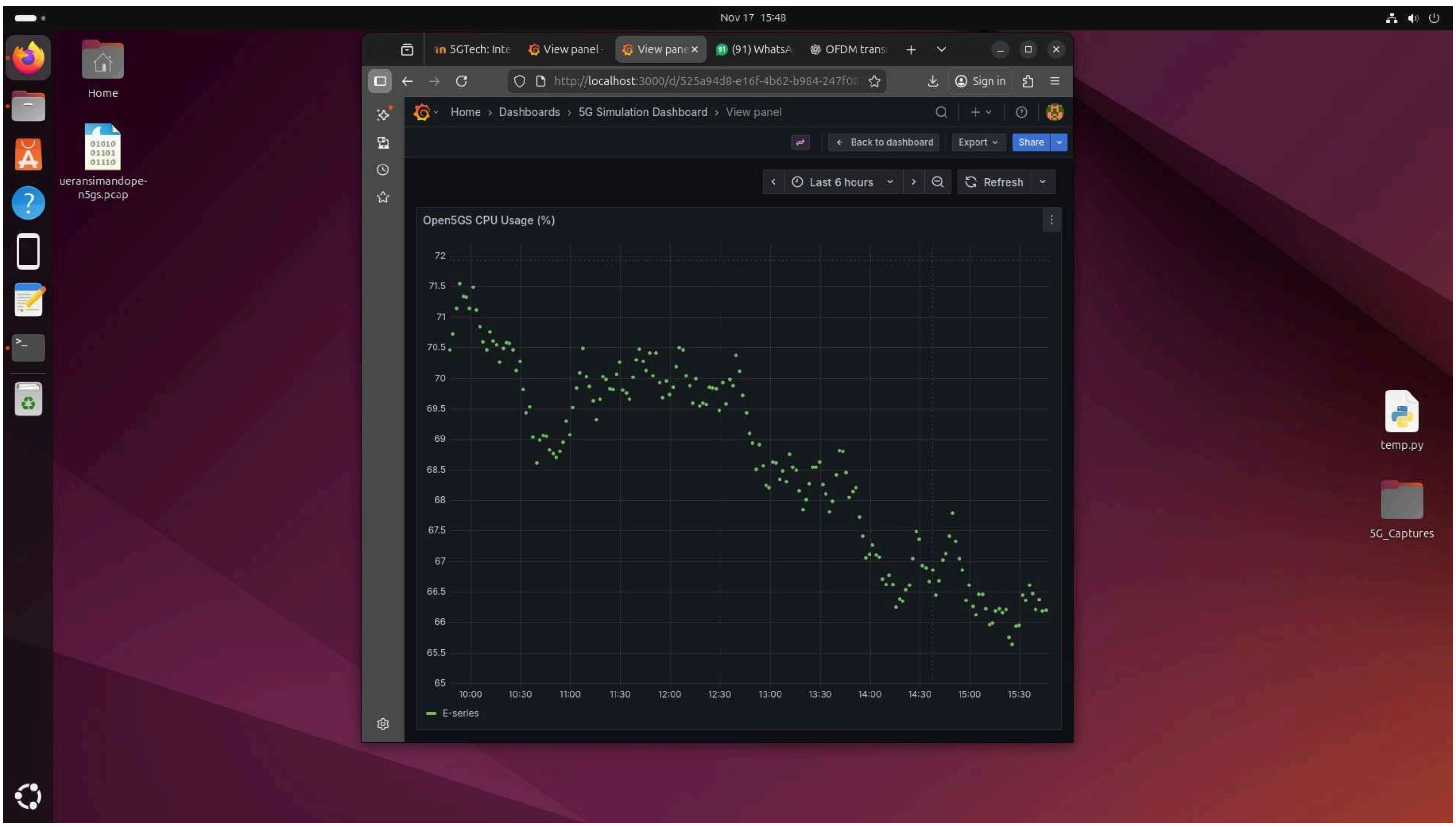
5G_Captures



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Back to dashboard
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Last 6 hours
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Open5GS CPU Usage (%)
72
71.5
71
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69.5
69
68.5
68
67.5
67
66.5
66
65.5
65
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